AI / Physics Applicability Map — evidence from this idealized 2D dataset

Prediction request: is it inside the sampled geometry / porosity domain?
First check physical assumptions, phase contrast and descriptor coverage.

Pure RF

Same generated families / porosity levels
10-split MAE = 1.657

Path descriptors supply the main gain.
RF and Hybrid are nearly tied here.
Small-data benefit of Hybrid not stable.

Hybrid (conditional)

Unseen middle porosity 0.60
MAE RF / Hybrid = 2.888 / 1.417

Smooth physical backbone helps here.
Residual must remain transferable.
Not a universal OOD improvement.

Physics (conditional)

High porosity, gas-dominated limit
Raw physics MAE = 0.125

3p shape physics best on family shift.
Porosity-only physics misses direction.
Need valid physics + shape assumptions.

Real aerogel outside this map: 3D / Knudsen / radiation / temperature / contact effects absent.
Unknown physics or distant geometry -> obtain new reference data; do not trust a leaderboard.